Assignment 8 — Task List, Milestones, Timeline, and Effort Matrix

Model Recommend (Concentrate AI)

Team: Allison Oh, Aditya Pawar

Covers Assignment #5 (Task Lists) and Assignment #6 (Milestones, Timeline, and Effort Matrix)

Contents: 1 Task List · 2 Milestones List · 3 Table 1: Timeline · 4 Table 2: Effort Matrix · 5 Work Breakdown Structure · 6 Risk Register · 7 Consistency Check

Senior Design 1, Fall 2026

Task List, Milestones, Timeline, and Effort Matrix

Project: Model Recommend (Concentrate AI) **Team:** Allison Oh, Aditya Pawar **Advisors:** Jillian Aurisano (Department of Computer Science), Todd Liebermann (Concentrate AI)

This package covers Assignment #5 (Task Lists) and Assignment #6 (Milestones, Timeline, and Effort Matrix). Every task uses the same ID, name and owner in the Task List, Table 1, Table 2, the work breakdown structure and the risk register. Module names come from *Project-Description.md* and *Design_Diagrams* D0–D2.

Calendar assumptions. Fall 2026 dates are the CS Senior Design I deadlines in Canvas. Spring 2027 (CS 5002) is assumed to run January 11 – April 30, with spring break March 15–21. We will move the internal milestones to the CS 5002 syllabus dates when it is published. No work is planned over winter break (December 12 – January 10). Both members are in classes, with no co-op, in Spring 2027.

1. Task List (Assignment #5)

The same list is committed as `Tasklist.md` in the repository root. Seven tasks are Allison's and eight are Aditya's (about five per member was the guide); each is sized to one or two weeks of work.

ID	Task	Owner
T1	Obtain the vendor API keys, the identity-provider tenant, and the managed PostgreSQL and container hosting from Concentrate AI for the Recommendation Server and Report Store.	Aditya Pawar
T2	Research the rate-limit headers, token-usage fields, prices and benchmarking terms of the OpenAI and Anthropic APIs for the Recommendation LLM adapter (I3).	Aditya Pawar
T3	Develop the Report Store schema, migrations and repository methods (Fig D2-1, I4) in PostgreSQL.	Allison Oh
T4	Develop the Configuration and Logging glue components (getConfig, structured logging, VendorCall usage log).	Allison Oh
T5	Develop the Authentication glue component (JWT verification, AppUser roles, requireRole) for the Recommendation Server.	Aditya Pawar
T6	Develop the Recommendation LLM adapter callVendor for OpenAI and Anthropic, with cost computed from token counts (I3).	Aditya Pawar
T7	Develop A1 parallel benchmark dispatch with per-vendor rate-limit handling and the rate_limit_warning in the Recommendation Server.	Aditya Pawar
T8	Develop A2 weighted scoring, ranking and the reasoning template in the Recommendation Server.	Aditya Pawar
T9	Develop the run lifecycle (Fig D2-3) and endpoints E1, E5 and E6 in the Recommendation Server.	Aditya Pawar
T10	Develop the Notifications glue component (outbox, dispatcher, approval reminder job; Fig D2-5).	Allison Oh

ID	Task	Owner
T11	Develop the vendor-approval, review and selection endpoints E2–E4 and E7–E12 in the Recommendation Server.	Allison Oh
T12	Develop the Recommendation Client brief form, run progress and shortlist screens S1–S3 (Fig D2-2).	Allison Oh
T13	Develop the Recommendation Client vendor-approval, review-queue and notification screens S4–S6.	Allison Oh
T14	Test the Recommendation Client and Recommendation Server end to end against AC-01.2, AC-02.1 and AC-02.2 and the edge cases EC-01 to EC-46.	Allison Oh
T15	Validate AC-01.1 (90-second shortlist) and AC-01.3 (rate-limit pause) under real vendor load and refine the A1 defaults.	Aditya Pawar

2. Milestones List (Assignment #6)

Course deliverables come first (dates from Canvas, all due 11:59 PM Eastern); internal milestones follow every one to three weeks through the implementation semester. The only gap longer than three weeks is winter break, when no work is planned (risk R4).

ID	Milestone	Date	Status	What it demonstrates	How it is verified	Verified by
A1	Professional Biography	Sep 2, 2026	**Complete**	Each member's background and goals	Submitted and graded in Canvas	Instructor
A2	Team Formation and Project Description	Sep 9, 2026	**Complete**	Team, advisors and topic agreed	Submitted and graded; Project-Description.md in repo	Instructor
A3	Advisor Selection and Design Constraints Essay	Sep 18, 2026	**Complete**	Advisors confirmed; economic, security, social and legal constraints	Submitted and graded; Constraints_Essay.md in repo	Instructor, advisors
A4	User Stories and Use Cases	Sep 28, 2026	**Complete**	US-01–US-05, UC-01–UC-02 and acceptance criteria	Submitted and graded; User_Stories.md in repo	Instructor
A5	High-Level System Design (D0)	Oct 2, 2026	**Complete**	Components, interfaces I1–I4, decision log	Submitted and graded; Design_Diagrams/D0 in repo	Instructor
A6	Detailed System Design, Part 1 (D1)	Oct 8, 2026	**Complete**	Data model, A1, A2, API E1–E4	Submitted and graded; Design_Diagrams/D1 in repo	Instructor
A7	Detailed System Design, Part 2 (D2) and Peer Review Notes	Oct 14, 2026	**Complete** (D2 in repo; peer review notes being finalized)	Every component has a data model, behavior and interface; edge cases, traceability, rationale	Design_Diagrams/D2 and Peer_Review_Notes.md in repo; peer review	Instructor, reviewing team
A8	Task List, Milestones, Timeline, and Effort Matrix	Oct 28, 2026	Complete on submission	This plan: owners, dates, effort and risks	Submitted PDF; Tasklist.md and Planning/ in repo	Instructor, advisors
A9	Requirements Document and Technical Specification	Oct 30, 2026	Planned	Requirements and specification; D2 traceability becomes the traceability matrix	Submitted and graded	Instructor

ID	Milestone	Date	Status	What it demonstrates	How it is verified	Verified by
A10	Test and Validation Plan Draft	Nov 4, 2026	Planned	How each acceptance criterion and EC row will be tested (input to T14, T15)	Submitted and graded	Instructor
A11	Ethics and Responsible-Use Statement	Nov 11, 2026	Planned	Data handling, vendor terms, explainability (constraints essay)	Submitted and graded	Instructor
A12	Presentation Draft and Documented Advisor Feedback	Nov 18, 2026	Planned	Presentation reviewed by advisors	Feedback recorded in the submission	Advisors, instructor
A13	Team Oral Presentation, Recorded Video, and Peer Reviews	Nov 18, 2026	Planned	Team can present the design and plan	Presentation delivered; peer reviews	Instructor, peers
A14	Design Report, Design Notebook, and Team and Self Assessment	Dec 2, 2026	Planned	Complete Fall design record; Week 14 report reproduces Tables 1 and 2	Submitted and graded	Instructor
A15	Project Advisor Assessments	Dec 8, 2026	Planned	Advisors' assessment of the team	Advisor forms submitted	Advisors
MI-1	Development environment ready	Dec 4, 2026	Planned	Keys, IdP tenant and hosting work (T1); vendor research memo with prices, headers and terms (T2)	A test call to each vendor succeeds from the hosted container; memo in repo	Aditya Pawar; Todd Liebermann confirms access
M0	CI and configuration deployed	Jan 20, 2027	Planned	CI runs tests on every push; config and logging live on the host (T4)	Green CI run; a startup log line in the platform	Allison Oh
M1	**Walking skeleton**	Jan 29, 2027	Planned	Authenticated request reaches the Server and reads and writes the Report Store (T3, T4, T5)	Demo: sign in, call a test endpoint, row written and read back	Both members; advisor review
M2	Real vendor benchmarks logged	Feb 16, 2027	Planned	A1 sends prompts to both vendors with per-vendor limits; every call in VendorCall with cost (T6, T7)	Run log shows calls, tokens and cost for 2 vendors; a forced 429 pauses one vendor	Aditya Pawar; Allison reviews
M3	Ranked shortlist through the API	Mar 9, 2027	Planned	E1 → run → E5 returns a scored, reasoned shortlist (US-01, US-02) (T8, T9)	Scripted API run on 3 briefs; scores 0–100; result in under 90 s	Both members
M4	**MVP advisor demo**	Mar 25, 2027	Planned	A product manager can submit a brief and read the shortlist in the Client (T12)	Live demo to advisors	Jillian Aurisano, Todd Liebermann
M5	Feature complete	Apr 10, 2027	Planned	Vendor approval, review, selection and notifications work (US-03, US-04, US-05) (T10, T11, T13)	Walk-through of UC-01 alternate flow and UC-02	Both members; advisor review
M6	**Validation complete**	Apr 28, 2027	Planned	All acceptance criteria pass; AC-01.1 met under real load (T14, T15)	Test report with pass/fail per AC and EC	Allison Oh (tests), Aditya Pawar (load); advisors sign off

ID	Milestone	Date	Status	What it demonstrates	How it is verified	Verified by
M7	**Final presentation and handoff to Concentrate AI**	Apr 30, 2027 (assumed)	Planned	Working system and documentation handed over	Final presentation; repository handed to Concentrate AI	Instructor, advisors

3. Table 1: Timeline (Assignment #6)

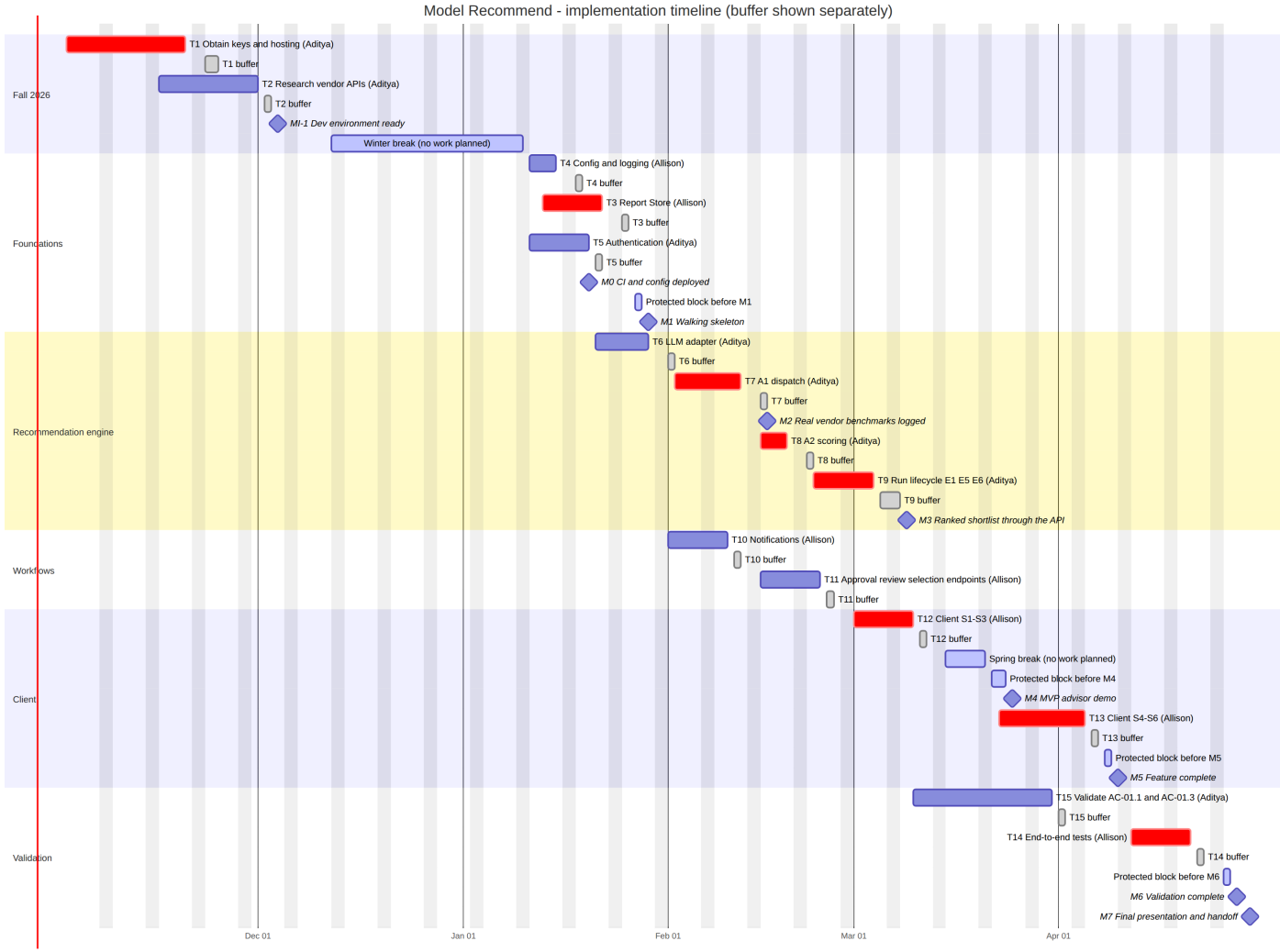
Each task's completion date includes its buffer, shown explicitly as working days and as a share of the task (15–20 percent). Major milestones (bold: M1, M4, M6, M7) and M5 have a protected block of 2–3 working days before them that holds no planned work and absorbs slips. **CP** marks the critical path: T1 → T3 → T7 → T8 → T9 → T12 → T13 → T14 → M6 → M7. Risk mitigations that take schedule time are noted in the last column.

ID	Task (short)	Owner	Start	Work done	Buffer	Completion	Depends on	CP	Risk notes
T1	Obtain keys, IdP and hosting	Aditya Pawar	Nov 2, 2026	Nov 20	3 d (17%)	Nov 25, 2026	none	✔	R2: request sent Nov 2; fallback by Nov 25
T2	Research vendor APIs, prices and terms	Aditya Pawar	Nov 16, 2026	Dec 1	2 d (15%)	Dec 3, 2026	none		R7: terms review here
MI-1	Development environment ready					**Dec 4, 2026**	T1, T2		
—	Winter break, no planned work		Dec 12, 2026			Jan 10, 2027			R4
T4	Configuration and Logging	Allison Oh	Jan 11, 2027	Jan 15	1 d (17%)	Jan 18, 2027	T1		
T3	Report Store schema and repository	Allison Oh	Jan 13, 2027	Jan 22	2 d (20%)	Jan 26, 2027	T1	✔	
T5	Authentication	Aditya Pawar	Jan 11, 2027	Jan 20	2 d (20%)	Jan 22, 2027	T1		
M0	CI and configuration deployed					**Jan 20, 2027**	T4		
—	Protected block before M1		Jan 27, 2027			Jan 28, 2027			
M1	**Walking skeleton**					**Jan 29, 2027**	T3, T4, T5	✔	
T6	Recommendation LLM adapter	Aditya Pawar	Jan 21, 2027	Jan 29	2 d (20%)	Feb 2, 2027	T2, T4		R3: mock vendor in CI
T10	Notifications	Allison Oh	Feb 1, 2027	Feb 10	2 d (20%)	Feb 12, 2027	T3, T4		

ID	Task (short)	Owner	Start	Work done	Buffer	Completion	Depends on	CP	Risk notes
T7	A1 parallel dispatch	Aditya Pawar	Feb 2, 2027	Feb 12	2 d (18%)	Feb 16, 2027	T3, T6	✔	R1
M2	Real vendor benchmarks logged					**Feb 16, 2027**	T6, T7	✔	
T8	A2 scoring and ranking	Aditya Pawar	Feb 15, 2027	Feb 19	1 d (17%)	Feb 22, 2027	T7	✔	
T11	Approval, review and selection endpoints	Allison Oh	Feb 15, 2027	Feb 24	2 d (20%)	Feb 26, 2027	T3, T5, T10		
T9	Run lifecycle, E1, E5, E6	Aditya Pawar	Feb 23, 2027	Mar 4	2 d (20%)	Mar 8, 2027	T7, T8	✔	
M3	Ranked shortlist through the API					**Mar 9, 2027**	T8, T9	✔	R1 check: run under 70 s
T12	Client S1–S3	Allison Oh	Mar 1, 2027	Mar 10	2 d (20%)	Mar 12, 2027	T5, T9 (mock API from Feb 22)	✔	R6: build against the D2 API contract first
T15	Validate AC-01.1 and AC-01.3	Aditya Pawar	Mar 10, 2027	Mar 31 (spring break excluded)	2 d (15%)	Apr 2, 2027	T9		R1: tune concurrency and deadline
—	Spring break (assumed), no planned work		Mar 15, 2027			Mar 21, 2027			R4
—	Protected block before M4		Mar 22, 2027			Mar 24, 2027			
M4	**MVP advisor demo**					**Mar 25, 2027**	T12	✔	
T13	Client S4–S6	Allison Oh	Mar 23, 2027	Apr 5	2 d (17%)	Apr 7, 2027	T11, T12	✔	
—	Protected block before M5		Apr 8, 2027			Apr 9, 2027			
M5	Feature complete					**Apr 10, 2027**	T10, T11, T13	✔	
T14	End-to-end tests	Allison Oh	Apr 12, 2027	Apr 21	2 d (20%)	Apr 23, 2027	T9, T13, T15	✔	
—	Protected block before M6		Apr 26, 2027			Apr 27, 2027			
M6	**Validation complete**					**Apr 28, 2027**	T14, T15	✔	
M7	**Final presentation and handoff**					**Apr 30, 2027**	M6	✔	Date assumed

Figure 1. Timeline (Gantt view of Table 1)

Model Recommend (Concentrate AI). Goal: recommend a particular AI model based on user input.



Conventions. Bars are tasks from the Task List (T-numbers) with their owner; red bars are on the critical path. Grey bars directly after a task are its buffer (15-20 percent of the task). Light blue bars are protected blocks before a milestone and periods with no planned work (winter break, assumed spring break). Diamonds are milestones from the Milestones List. Weekends are shaded and excluded from durations.

4. Table 2: Effort Matrix (Assignment #6)

Each estimate is the PERT expected value (O + 4M + P) / 6 from the owner's optimistic, most likely and pessimistic hours. The owner (**bold**, ★) carries the largest share of every task; the other member's hours are code review, pairing on interfaces, and testing.

ID	Task (short)	O / M / P (h)	Expected (h)	Aditya Pawar (h)	Allison Oh (h)	Row total (h)
T1	Obtain keys, IdP and hosting	4 / 8 / 16	9	**8 ★**	1	9
T2	Research vendor APIs, prices and terms	4 / 6 / 10	6	**5 ★**	1	6
T3	Report Store schema and repository	12 / 18 / 30	19	4	**15 ★**	19
T4	Configuration and Logging	6 / 10 / 16	10	2	**8 ★**	10
T5	Authentication	8 / 12 / 20	13	**11 ★**	2	13
T6	Recommendation LLM adapter	10 / 16 / 26	17	**15 ★**	2	17
T7	A1 parallel dispatch	16 / 24 / 40	25	**21 ★**	4	25
T8	A2 scoring and ranking	8 / 12 / 20	13	**10 ★**	3	13
T9	Run lifecycle, E1, E5, E6	10 / 16 / 26	17	**13 ★**	4	17
T10	Notifications	10 / 14 / 24	15	3	**12 ★**	15
T11	Approval, review and selection endpoints	12 / 18 / 28	19	3	**16 ★**	19
T12	Client S1–S3	14 / 20 / 32	21	4	**17 ★**	21
T13	Client S4–S6	14 / 20 / 32	21	3	**18 ★**	21
T14	End-to-end tests	10 / 16 / 26	17	4	**13 ★**	17
T15	Validate AC-01.1 and AC-01.3	8 / 14 / 24	15	**12 ★**	3	15
	Column total (h)		**237**	**118 (49.8%)**	**119 (50.2%)**	**237**
	Tasks owned			8	7	15

Balance. The hour columns are even (118 vs 119), but Aditya owns eight tasks to Allison's seven, and his tasks sit on the critical path in February while Allison's load peaks in March–April. Aditya's hours are front-loaded (95 h before M3, 23 h after), while Allison's are more even but heavier late in the term (68 h before M3, 51 h after).

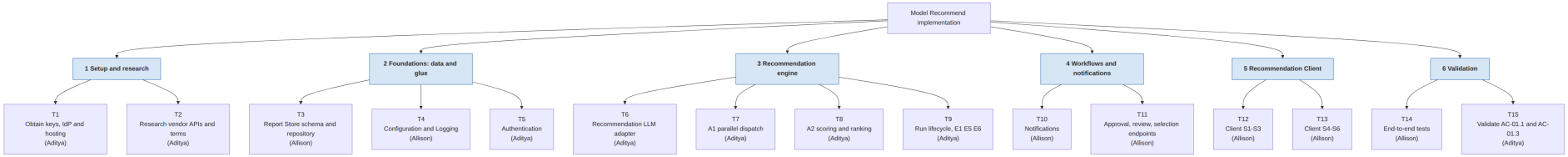
Rebalancing plan. Each member logs hours weekly in the Planning folder. At each milestone we compare logged hours with this matrix; if either member is more than 10 percent over plan, the next unstarted task moves its secondary hours to the lighter member, or the owner changes (first candidates: T11 or T15, which either member can do). After M3, Aditya takes over the Playwright test scaffolding inside T14 (raising his share there) so the April load is shared.

5. Work Breakdown Structure

Phase (level 1)	Tasks (level 2)
1 Setup and research	T1 Obtain keys, IdP and hosting (Aditya); T2 Research vendor APIs, prices and terms (Aditya)
2 Foundations: data and glue	T3 Report Store schema and repository (Allison); T4 Configuration and Logging (Allison); T5 Authentication (Aditya)
3 Recommendation engine	T6 Recommendation LLM adapter (Aditya); T7 A1 parallel dispatch (Aditya); T8 A2 scoring and ranking (Aditya); T9 Run lifecycle, E1, E5, E6 (Aditya)
4 Workflows and notifications	T10 Notifications (Allison); T11 Approval, review and selection endpoints (Allison)
5 Recommendation Client	T12 Client S1–S3 (Allison); T13 Client S4–S6 (Allison)
6 Validation	T14 End-to-end tests (Allison); T15 Validate AC-01.1 and AC-01.3 (Aditya)

Figure 2. Work breakdown structure

Model Recommend (Concentrate AI). Goal: recommend a particular AI model based on user input.



Conventions. Level 1 (blue boxes) are the six implementation phases; level 2 are the Task List items with the same IDs and owners as Table 1 and Table 2. Lines show which phase each task belongs to.

6. Risk Register

Likelihood and impact are Low, Medium or High. Each risk names where its mitigation shows up in Table 1.

ID	Category	Risk	Likelihood	Impact	Strategy	Mitigation (where it shows in Table 1)	Owner	Trigger
R1	Technical	Vendor latency and rate limits make the 90-second shortlist (AC-01.1) unreachable	Medium	High	Reduce	T2 measures limits early; M3 checks run time; T15 tunes concurrency and the deadline before M4 and M6	Aditya Pawar	A run at M2 or M3 takes over 70 s
R2	External	Concentrate AI is slow to provide API keys, the identity-provider tenant or hosting	Medium	High	Reduce	T1 starts Nov 2 with 3 d buffer; fallback to team-paid keys, local Docker PostgreSQL and a free IdP tier so T3–T5 can start Jan 11	Aditya Pawar	Access not granted by Nov 25
R3	External	Vendor spend during development grows past what Concentrate AI expects	Medium	Medium	Reduce	Mock vendor in CI (T6); per-client run caps; weekly spend check from the VendorCall log	Aditya Pawar	Spend above USD 50 in a week
R4	Team	Finals, winter break and spring break cut available hours	High	Medium	Accept	No tasks planned Dec 12 – Jan 10 or Mar 15–21; buffer on every task	Allison Oh	Any task slips more than 3 working days
R5	Team	One member is unavailable for a week or more (two-person team)	Low	High	Reduce	Secondary hours on every task (Table 2) so each member knows the other's code; shared docs	Allison Oh	A member is out more than 5 working days
R6	Technical	The Client is blocked waiting for the Server API	Medium	Medium	Reduce	T12 builds against a mock of the D2 API contract from Feb 22; T9 is on the critical path and M3 checks it	Allison Oh	T9 not complete by Mar 8
R7	External	A vendor's terms restrict benchmarking or showing comparisons	Low	High	Avoid	T2 reviews terms; reports stay private to each client (EC-35); no public comparisons	Aditya Pawar	A restrictive clause is found in T2

ID	Category	Risk	Likelihood	Impact	Strategy	Mitigation (where it shows in Table 1)	Owner	Trigger
R8	Technical	The identity provider choice (D2 gap G8) delays authentication	Medium	Medium	Reduce	Decided in T1; T5 uses standard OIDC so the provider can change	Aditya Pawar	No provider chosen by Dec 4 (MI-1)

7. Consistency Check

ID	Task List	WBS phase	Table 1	Table 2	Owner in all four
T1–T2	✔	1	✔	✔	Aditya Pawar
T3–T4	✔	2	✔	✔	Allison Oh
T5	✔	2	✔	✔	Aditya Pawar
T6–T9	✔	3	✔	✔	Aditya Pawar
T10–T11	✔	4	✔	✔	Allison Oh
T12–T13	✔	5	✔	✔	Allison Oh
T14	✔	6	✔	✔	Allison Oh
T15	✔	6	✔	✔	Aditya Pawar